

Response to the Reviewer

Manuscript: *Corner-Speed-Based Design and Optimization of a Switched Reluctance Motor Drive for Low-Voltage Electric Bicycles*

(previous title: *Analysis and Design Considerations for a Switched Reluctance Motor-Based Drive System for an Electric Bike*)

We thank the reviewer for a detailed and constructive review. We agree with the main criticisms, and the revision substantially extends the work. The largest changes are as follows.

1. **FE model.** A nonlinear 2-D finite-element (FE) model of the full machine was added (Section IV-B). It includes a mesh-convergence study and a quantitative comparison with the analytical model at 8, 15, 23 and 31 A (Table VI, Fig. 3).
2. **FE-based results.** All drive-level results are now computed with the FE flux-linkage characteristic instead of the analytical one.
3. **TSF control.** Torque-sharing-function control (linear, cubic and exponential) was implemented and compared with current-reference control (Section VII).
4. **Sensitivity studies.** New studies cover PWM frequency, a four-node transient thermal network, the gear model, DC-link current sharing, air gap, pole arcs, the dwell limit and the convergence of the angle search.
5. **Novelty.** A new subsection (II-E) positions $P_c = k_c V_{dc} I_{pk}$ relative to the classical volt-second and converter volt-ampere arguments.

Experimental validation has not been added. We state this explicitly in the abstract, the introduction and Section X-D, and we reframed the paper as an analytical, FE and simulation-based design methodology (see comment 5).

Because the FE characteristic and a more thorough angle search replaced the analytical model, several headline numbers changed. Table 1 below lists them. Line-by-line changes are given under each comment.

Quantity	Original manuscript	Revised (FE-based)
Current for launch torque / converter limit	27.9 A / 30.7 A	26.5 A / 29.2 A
Launch torque at limit (margin)	6.81 N·m (+14.9%)	6.63 N·m (+11.6%)
Rated η_{mot} / η_{sys}	75.8 / 72.8%	80.2 / 77.9%
Peak η_{mot} / η_{sys} (map)	81.4 / 78.6%	80.6 / 78.3%
Torque ripple, launch / rated (current-ref.)	113 / 148%	60 / 87%
Route consumption / estimated range	5.05 Wh/km / 64 km	5.12 Wh/km / 63 km

Table 1: Main numerical changes. The rated-efficiency increase comes mostly from the improved angle search (comment 26).

Reviewer comment 2–3 (strengths; novelty of the corner-speed argument)

The reviewer asks us to establish whether $P_c = k_c V_{dc} I_{pk}$ is new or a reformulation of established volt-second and energy relationships.

Response. We agree that the relation is built from established ingredients. The volt-second limit is standard. The energy-conversion-loop argument that links converter volt-amperes to output power through the ψ - i loop utilization is classical (Miller’s energy ratio and converter volt-ampere requirement). We no longer present the relation as a new law. The contribution is now stated as how the relation is *used* for low-voltage sizing:

- choosing the turns at a mission-derived corner speed rather than at the speed limit;
- showing that the converter current is fixed before the machine is sized, so that enlarging the machine buys efficiency, not current;
- reporting k_c for this machine class.

Changes. New Section II-E, *Relationship to Existing Volt-Second and Energy-Based Methods*, as the reviewer suggested. Contribution 1 in the introduction and the conclusion are reworded accordingly. The trade study confirms the claim quantitatively: I_{pk} varies by 22% over a fourfold range of mass, with $k_c = 0.54$ – 0.66 across nine designs.

Reviewer comment 4 (electromagnetic model insufficiently validated; FE required)

Response. Done. We implemented a nonlinear 2-D magnetostatic FE model of the full 12/8 cross-section with the following features:

- vector potential formulation, first-order triangles and Newton iteration;
- M270-35A B-H curve and re-meshing at every rotor position;
- Arkkio torque, cross-checked against the co-energy derivative to within 0.5%.

Mesh convergence is shown in Table V: halving the gap element size changes results by less than 0.1%. As requested, Table VI and Fig. 3 compare the analytical model and FE at 8, 15, 23 and 31 A:

- flux linkage: RMS error 13–15%, maximum 23%;
- aligned inductance: +5.7%; unaligned inductance: −9.7%;
- mean torque: −2 to −17%; peak torque: within $\pm 10\%$.

Because these errors are significant, *all drive-level results now use the FE characteristic*. It is imported as $\psi(i, \theta)$ tables, with torque computed from the co-energy of the same table so that the model stays energy-consistent. Fig. 4 shows the FE field solution.

Changes. Sections IV-B and IV-C, Tables V and VI, Figs. 3 and 4. Sections VI–VIII were recomputed with the FE characteristic.

Reviewer comment 5 (no experimental prototype)

Response. We fully agree that experimental validation would strengthen the paper. It is not available for this revision, and we have not presented any simulated quantity as measured. Following the reviewer’s alternative recommendation, the paper has been reframed.

- The title now emphasizes the design methodology.
- The abstract states that “No prototype measurements are reported, and all performance figures are model predictions”.
- The introduction limits the scope explicitly.
- Section X-D lists the planned dynamometer tests (phase voltage and current, torque and ripple, DC input power, efficiency map, temperature rise, acoustic noise).

Changes. Title, abstract, Section I (scope paragraph), Section X-D and the conclusion. *[Note to authors: if prototype data become available before resubmission, Section X-D is where they would replace the stated limitation.]*

Reviewer comment 6 (torque ripple; TSF only discussed, not implemented)

Response. Done. We implemented linear, cubic and exponential TSFs with current references obtained from the inverse FE torque characteristic and tracked by a three-level hysteresis controller. Table XII and Fig. 11 compare them with current-reference control (CCC), in the format the reviewer proposed:

- At 200 r/min, cubic TSF reduces ripple from 197% to 10%.
- At 500 r/min it reaches 33%, and at 800 r/min and 1.5 N·m, 39% (against 83% for CCC).
- The cost is 12–28% higher RMS current and 7–12 points of efficiency.
- Above about 800 r/min, no TSF solution exists within I_{max} at 36 V.

This is the known speed limit of TSF, and it is particularly restrictive at low voltage. We therefore recommend a hybrid controller: TSF for launch, walk-assist and low-speed climbing, and angle control above about 12 km/h. The FE-based current-reference ripple is itself lower than originally predicted (87% at the rated point instead of 148%).

Changes. New Section VII, Table XII, Fig. 11, and the abstract and conclusion.

Reviewer comment 7 (launch margin should be stated)

Response. Done. The FE-based capability at the current limit is 6.63 N·m, which exceeds the 5.93-N·m requirement by 11.6%. The original analytical figure was 6.81 N·m (+14.9%). The gear-sensitivity study shows that the margin shrinks to 6% if $\eta_g = 0.90$.

Changes. Section VI-A, abstract, Table XIII.

Reviewer comment 8 (efficiency interpretation; comparison with PM motors)

Response. We agree that the comparison was not like-for-like. We removed the quantitative comparison with commercial PM hubs, and Section X-C now states why a fair comparison requires identical boundaries, voltage, temperature and method. Efficiency boundaries are defined explicitly in Section IV-F (η_{mot} , η_{sys} and η_{conv} , with

the gear excluded, so that battery-to-wheel efficiency equals $\eta_{sys}\eta_g$). The Fig. 8 caption states the boundary.

Changes. Sections IV-F and X-C, Fig. 8 caption.

Reviewer comment 9 (thermal model too simplified; sensitivity)

Response. Done. The one-node model was replaced by a transient four-node network (winding, stator, rotor, shell) with temperature-dependent copper loss. Table XI gives the sensitivity the reviewer requested: $h = 15/25/40$ W/m²K, ambient 35/45 °C (25 °C is also computed), continuous 250 W, 6% climb, and repeated launches (5 s every 60 s). Fig. 10 shows temperature versus time. One new finding is that a sustained 6% climb at $h = 15$ W/m²K and 45 °C reaches 142 °C at steady state. We therefore now recommend class-H insulation or thermal derating for hot or mountainous use, instead of claiming a wide margin.

Changes. Section VI-F, Table XI, Fig. 10.

Reviewer comment 10 (DC-link capacitor derivation)

Response. Done. Section VI-E now gives the charge bound explicitly as Eq. (16) and explains that it assumes the battery supplies only the mean current. It gives 1.56 mF for 5% ripple at the rated point, but a meaningless 30 mF at launch, where the stroke frequency is only 32 Hz. We therefore added a harmonic current-sharing analysis between the capacitor ($R_{ESR} = 20$ m Ω) and the battery ($R_b = 0.15$ Ω , $L_b = 1$ μ H), with all assumptions stated (Table X). The conclusions are:

- 1 mF gives 4.8% ripple at the rated point;
- the launch ripple is set by battery impedance;
- the bank is governed by ripple current (13.4 A RMS), not capacitance.

The original 3.8-mF recommendation is withdrawn in favor of 1–2.2 mF rated for at least 14 A RMS.

Changes. Section VI-E, Eq. (16), Table X.

Reviewer comment 11 (switching frequency; production PWM controller)

Response. Done. We implemented fixed-frequency peak-current-mode PWM in the simulator and accumulated switching energy per transition at the actual current, including diode reverse recovery. The hill, rated and launch points were evaluated at 4, 8, 16 and 20 kHz, as requested (Table IX). At 36 V, switching loss is at most 1.9 W (launch, 20 kHz), and η_{sys} changes by less than 0.1 point, so the efficiency results also hold for the recommended production PWM.

Changes. Section VI-D, Table IX.

Reviewer comment 12 (gearbox efficiency assumption)

Response. Done. The gear model now includes an optional speed-proportional drag loss in addition to the mesh efficiency (Eq. (2)). Table XIII reports the sensitivity to $\eta_g = 0.90, 0.93, 0.95$ and 0.97 and to two drag cases. Consumption ranges from 5.02 to 5.38 Wh/km, and the launch requirement ranges from 5.81 to 6.26 N·m. The launch requirement stays within the envelope in all cases.

Changes. Section II-B, Eq. (2), Section VIII, Table XIII.

Reviewer comment 13 (route model quasi-static; braking assumption)

Response. We now list the neglected effects explicitly: transients, battery sag, controller delay and thermal state. Battery sag is partly covered by design, because the turns are chosen at the 30-V end-of-discharge voltage. The mechanical-braking assumption is now justified quantitatively. The brakes dissipate 7.2 Wh on the route, so regeneration could return at most about 5 Wh (about 15% of battery energy on this hilly route). The asymmetric bridge supports regeneration without hardware change.

Changes. Section VIII.

Reviewer comment 14 (64-km range claim)

Response. Changed to “an *estimated range of about 63 km under these route and modeling assumptions*” everywhere, with the gear sensitivity (60–65 km).

Changes. Abstract, Section VIII, Table XIII, conclusion.

Reviewer comment 15 (position sensing details)

Response. Expanded. The text now covers extrapolation with speed and acceleration, quantization and sensor alignment ($\pm 1^\circ$) as commutation-angle errors, the stand-still hill start (exciting the phase pair that gives positive torque throughout the 15° Hall sector), and direction reversal.

Changes. Section X-A.

Reviewer comment 16 (mechanical feasibility; tolerance stack-up)

Response. Done in two parts. First, FE air-gap sensitivity (Table XV): at 0.35 mm the 30-A mean torque drops by 4.0% (8.3% at 15 A), and at 0.40 mm by 8.2% (16%). Second, a tolerance allocation in the format the reviewer suggested (Table XVI), clearly labeled as an assumed design budget rather than a measurement. It gives a minimum local gap of 0.215 mm worst case and 0.26 mm by RSS, with the corresponding eccentricity and unbalanced magnetic pull.

Changes. Section X-B, Tables XV and XVI.

Reviewer comment 17 (mutual coupling neglected)

Response. Quantified by FE with two phases excited simultaneously (Section IV-D). In the commutation overlap, the second phase changes the flux linkage by at most 7.2%, and the torque differs from superposition by at most 0.36 N·m (6.9%). The mutual coupling is not yet included in the drive model. We state this, and we note that it bounds the achievable TSF ripple without online compensation.

Changes. Section IV-D, Sections VII and X-D.

Reviewer comment 18 (literature review needs 2020–2026 references)

Response. Added verified recent references:

- Terzić and Mihić (2022), an SRM design for a mid-drive e-bike, the closest prior work, which the paper now positions against;
- Lan *et al.* (2021), a review of SRM EV powertrains;
- Fang *et al.* (2021), a review of SRM control and vibration;
- Abdalmagid *et al.* (2022), geometry and topology optimization;
- Hamouda *et al.* (2020), torque control for light EVs;
- Bostanci *et al.* (IEEE TTE, 2017);
- Xue *et al.* (2009), TSF optimization.

[Note to authors: please add further 2023–2026 papers you consider essential, e.g. on SRM acoustic noise and integrated hub motors; we only included references whose bibliographic details we could verify.]

Reviewer comment 19 (FE/experimental plots)

Response. FE plots were added: the validation figure (Fig. 3) and the field plot (Fig. 4). All performance figures (Figs. 6–9 and 12) are now FE-based. No experimental plots are available; see comment 5.

Reviewer comment 20 (title)

Response. Adopted the reviewer’s preferred title: *Corner-Speed-Based Design and Optimization of a Switched Reluctance Motor Drive for Low-Voltage Electric Bicycles*.

Reviewer comment 21 (summary of major comments)

Response. Major comments 1–7 are addressed as described under comments 4–5 (validation), 6 (torque ripple), 2–3 (novelty), 9 and 4 (loss and thermal models), 11 (controller mismatch), 12 (gear) and 14 (range).

Reviewer comment 23 (“halves the launch current” phrasing)

Response. Rephrased as the reviewer suggested. Section II-D and the conclusion now read: “For the specified 36-V, 500-W corner power, this approximately halves the peak current relative to sizing the turns for launch torque at 25 km/h”. The assumptions (one-stroke dwell, fixed V_{dc} , similar k_c) are stated next to the claim.

Reviewer comment 25 (minor corrections)

- *Placeholder authors.* These must be filled in by the authors; they are marked TODO in the source.

- *Mass definition.* 3.4 kg is now defined as the active electromagnetic mass (laminations and copper), excluding housing, shaft, bearings and gear (Section III-D, Table IV).
- *Efficiency definitions and Fig. 7 boundary.* See comment 8 (Section IV-F, Fig. 8 caption).
- *1.1% power-balance error.* With the FE tables it is 1.4%. It is now explained as the error of the explicit time integration and the bilinear table interpolation (Section IV-E).
- *Repository link.* The repository URL is given in the first-page footnote and as reference [20].
- *Optimization grid and convergence.* Full details are in Section V-A, together with a convergence check against a $4\times$ denser grid: η_{sys} improves by at most 1.5 points and maximum torque by at most 2.9%.
- *Why $0.55\tau_r$.* A dwell sweep shows that $0.55\tau_r$ maximizes base-speed torque (2.07, 2.76, 3.50, 3.95 and 3.77 N·m for 0.40–0.60 and above). The original text also wrongly stated that the limit avoids continuous conduction. The optimum above base speed in fact runs in mild continuous conduction (minimum current 1.3–1.7 A), and this is now stated (Section V-A).
- *Pole-arc sensitivity.* Six FE pole-arc variants are reported in Table III.
- *Uncertainty.* Model-uncertainty statements are collected in Section X-D. Error bars are not meaningful without measurements, but the FE-versus-analytical differences (Table VI) and the grid-convergence bounds quantify the model uncertainty.

Reviewer comment 26 (additional change made by the authors)

Response. While addressing the dwell-limit question, we found that the original angle search was too coarse. A tighter dwell limit produced a *better* rated point. We therefore added dwell-based turn-on candidates to the search and verified convergence (comment 25). This, more than the change to the FE characteristic, explains why the rated efficiency increased from 72.8% to 77.9%. The trade study was recomputed with the improved search. It no longer supports the original claim that 11:1 gearing gains about four points at equal mass: at similar mass, 8:1 and 11:1 are within one point. Section IX has been corrected accordingly.

All code, FE scripts and data needed to reproduce every table and figure are in the public repository cited in the manuscript.